# Project Proposal: CareBridge Khulna

Team: CSE4104-7B-T01 | Section: 7B | CSE 4104: Software Development III
GitHub: https://github.com/114nahid/cse4104-7b-t01-carebridge-khulna

## 1. Project Title

**CareBridge Khulna: An AI-Powered Healthcare Assistance Platform for Elderly People**

CareBridge Khulna is a web application for elderly people living in Khulna city and for their sons and daughters who live far away from them. Using this system, a family member can find the right doctor for their parents, book a verified university student who will take them to the hospital and bring them back home, and see what is happening during the visit from anywhere. We will also use AI to make the process easier, mainly for people who are not very comfortable with technology.

## 2. Problem Statement

In Bangladesh, it is now very common for children to live away from their parents. Many young people from Khulna work in Dhaka or other cities, and a lot of them work abroad in countries like Saudi Arabia, Malaysia, the UK or the USA. Their parents usually stay in Khulna. As they get older, these parents need regular checkups for diabetes, high blood pressure, heart disease and similar problems. But going to the doctor is not easy for them. They have to find transport, wait for the serial, do tests, collect reports and then understand what the doctor said. Doing all of this alone is tiring for an elderly person and sometimes it is not safe either.

The children who live far away feel this problem the most. They want their parents to see a doctor when needed, but they have nobody to send with them. Sometimes a relative or neighbour helps, but not every time. Because of this, many elderly people delay their treatment, miss follow-up visits or go alone with a lot of trouble. After the visit, the family usually gets a short phone call and does not clearly know what the doctor said or which medicines were given.

There are some websites and apps in Bangladesh where people can search for doctors and book appointments. But these services stop after booking. Nobody goes with the patient, nobody helps them inside the hospital and the family gets no update. Families also often do not know which specialist to choose for a problem, and prescriptions are hard to read. So the main problem, taking the elderly person to the doctor safely and keeping the family informed, is still not solved.

On the other hand, there are thousands of university students in Khulna. Many of them come from middle-class families and look for part-time work to cover their own expenses. They are educated, they know the city and they can handle hospital work easily. If they are properly

verified, they can be the right people to go with elderly patients. Students get a good way to earn money, and families get a person they can trust.

AI can make this system much more useful. A family member can write the problem in Bangla, for example "my father feels pressure in his chest when he walks", and AI can suggest which type of specialist he should see. After the visit, AI can read the prescription photo and explain the medicines and their timing in simple Bangla. It can also turn the student's short notes into a proper report for the family, and it can catch emergency words like "fainted" or "severe chest pain". These things are very hard to do with normal if-else programming, but AI can do them well.

What makes our idea different is that it does not stop at finding a doctor. It covers the full visit, from home to hospital and back home, with a real person beside the patient and live updates for the family. As far as we know, no service in Khulna currently offers this together with AI support in Bangla.

## 3. Objectives

The main objectives of our project are:

1. To help families, especially those living abroad or in other cities, arrange medical care for their elderly parents in Khulna from anywhere.
2. To make sure that elderly patients do not have to go to the hospital alone, by providing verified university students as hospital assistants.
3. To create a safe and organized part-time earning opportunity for university students.
4. To keep family members informed at every step of the hospital visit.
5. To use AI for suggesting the right specialist, explaining prescriptions in simple Bangla and preparing visit reports.
6. To build a simple system that works on both mobile phones and computers, so that anyone can use it easily.

## 4. Proposed Features

The system will have three types of users: family members, student assistants and the admin. Each user will log in with their own account and will get their own dashboard. The features for each user are given below.

### 4.1 Features for Family Members

- **Account:** A family member can sign up with name, phone number, email and password, and then log in to their dashboard.

- **Patient profile:** They can add one or more elderly parents with age, gender, address, phone number and old health problems such as diabetes or blood pressure.
- **Home location on map:** They can pin the patient's house on a map, so the student can find the house without calling again and again.
- **Doctor search:** They can search doctors by specialty (for example, cardiologist or eye specialist) and by area. Each doctor will show the chamber address, visiting days and fee.
- **Search by problem:** If they do not know which doctor to choose, they can write or speak the problem in Bangla or English, and the AI will suggest the right type of specialist.
- **Book a hospital assistant:** After choosing a doctor and a date, they can book a student assistant for the visit. They can also cancel the booking before it is confirmed.
- **Visit tracking:** During the visit, they will see the status change step by step: picked up, reached hospital, with doctor, tests done and returned home.
- **After the visit:** They can see the prescription photo, the AI prescription summary and the visit report, and all old visits will stay saved in the history.
- **Rating and review:** At the end, they can give the student a rating from 1 to 5 and write a short review.

### 4.2 Features for Student Assistants

- **Registration:** A student will sign up with their university name, student ID, phone number, a photo of the university ID card and a photo of the NID.
- **Verification:** The student can start working only after the admin checks the documents and approves the account.
- **Available time:** The student can set which days and times they are free, so the admin only assigns visits at those times.
- **Assigned visits:** The student will see the visits given to them with the patient's name, address, map location, doctor and time, and can accept or decline them.
- **Status update:** During the visit, the student will update each step from their phone with one tap, so the family can follow along.
- **Prescription upload and notes:** After meeting the doctor, the student will upload a photo of the prescription and write a few short notes about what the doctor said.
- **Work history:** The student can see all completed visits, ratings and reviews in one place.

### 4.3 Features for Admin

- **Student verification:** The admin will check student documents and approve or reject new students.

- **Doctor management:** The admin will add, edit and remove doctor information such as specialty, chamber, visiting days and fee.
- **Booking management:** The admin will see new bookings, confirm the doctor's serial by phone and assign a free student to each booking.
- **Monitoring:** The admin can see all bookings with their current status, and can check ratings and complaints. A student with repeated bad reviews can be blocked.
- **Simple reports:** The admin will see total bookings, completed visits and active students on the dashboard.

### 4.4 AI Features

We will use the Gemini API for the AI part. The AI features are:

1. **AI Specialist Finder:** The user describes the patient's problem in Bangla or English, by typing or by voice. The AI understands it and suggests which type of specialist to see, with a short reason. For example, for "my mother has knee pain and cannot climb stairs", it may suggest an orthopedic doctor. The system will then show doctors of that type.
2. **AI Prescription Summary:** Handwritten prescriptions are hard to read for most families. From the prescription photo uploaded by the student, the AI will read the medicine names and explain in simple Bangla which medicine to take, how many times a day, before or after meals, for how many days, and when the next visit is.
3. **AI Visit Report:** The student's notes are usually short and informal. The AI will turn them into a clean report with the date, doctor, main advice, tests given and next steps. The family can read it in Bangla or English.
4. **Emergency Alert:** If the user writes or says something serious, like "fainted", "cannot breathe" or "severe chest pain", the system will not continue with a normal search. It will show the 999 number and nearby emergency departments right away.

The AI will not diagnose any disease and will not suggest any medicine by itself. It will only suggest the type of doctor and explain what a real doctor has already written. A short note about this will be shown on every AI result.

### 4.5 Other Features

- Notifications for new bookings, assignments and every status change.
- Secure login with different permissions for each type of user.
- Bangla and English text in the main parts of the website.
- A responsive design that works properly on mobile phones.

## 5. Technology Stack

We have chosen the following technologies for our project:

| Part | Technology | Purpose |
|---|---|---|
| Frontend | React.js, Tailwind CSS | User interface and dashboards |
| Backend | Node.js, Express.js | REST APIs and business logic |
| Database | PostgreSQL | Storing users, doctors, bookings and reviews |
| Authentication | JWT, bcrypt | Secure login and password hashing |
| AI Integration | Gemini API | Specialist finder, prescription summary, visit report |
| Voice Search | Web Speech API | Bangla voice input |
| Image Storage | Cloudinary | Prescription photos and student documents |
| Map | Leaflet, OpenStreetMap, OpenRouteService | Home location, chamber location and distance |
| Deployment | Vercel, Render, Supabase | Hosting the frontend, backend and database |
| Other Tools | GitHub, Figma, Postman, Draw.io | Version control, UI design, API testing and diagrams |

## 6. Expected Outcome

With CareBridge Khulna, a son or daughter living in Dubai or Dhaka will not feel helpless when their parents need a doctor. They can arrange everything from their phone, knowing that a trusted person is with their parents and seeing what is happening at every step. After the visit, they will get a clear written report instead of an unclear phone call.

Elderly patients will not have to go to the hospital alone. Having someone with them will make the visit safer and less stressful, and it will also encourage them to go for regular checkups, so that problems can be found early.

University students will get a chance to earn money while doing something good for society. Since every student will be verified and rated by families, good students will build trust over time and get more work.

The AI features will save time in finding the right doctor, make prescriptions easy to understand and give families a proper report without extra work for the student. In an emergency, the system will help people act faster. At the end of the semester, we expect to have a working website, deployed online, that real families in Khulna can use.

## 7. How the System Works

Let us take a simple example. Rahim works in Dubai, and his mother lives in Sonadanga, Khulna. She needs to see a heart specialist, but Rahim cannot come home. He opens CareBridge Khulna and writes in Bangla that his mother feels chest pressure while walking. The AI suggests that she should see a cardiologist. Rahim chooses a cardiologist from the list, picks a date and books a hospital assistant.

The admin confirms the serial by phone and assigns a verified student named Karim. On the visit day, Karim goes to the house and updates the status to "picked up", and Rahim sees it in Dubai. Then Karim updates "reached hospital" and "with doctor". After the checkup, Karim uploads a photo of the prescription and writes a few notes. The AI explains the prescription in simple Bangla and prepares a short visit report. When Karim brings Rahim's mother back home and updates "returned home", Rahim already knows everything. At the end, he gives Karim a rating.

From the technical side, the React frontend will talk to the Express backend through REST APIs. The backend will check the login with JWT, save and read data from the PostgreSQL database, send requests to the Gemini API for the AI features, store images in Cloudinary and use OpenStreetMap for the map.

## 8. Project Scope

In this semester, we will build the complete system for the three types of users, the doctor directory, hospital assistant booking, student verification, step-by-step visit status, the four AI features, ratings and notifications. The website will be deployed online so that it can be used from anywhere.

To keep the project realistic for one semester, we will not build live GPS tracking, online payment, a direct connection with hospital systems, a separate login for doctors or a separate mobile app. The serial will be confirmed by phone, payment will be made in cash, and the website itself will work on mobile phones. These features can be added in the future.

## 9. Initial Project Plan

We have divided our work over the semester as follows:

| Week | Work | Deliverable |
|---|---|---|
| 1 | Team formation and idea selection | Team information and project idea |
| 2 | Project proposal | Proposal and GitHub repository |
| 3 | Requirement analysis | SRS document |
| 4 | System design | Architecture, ER diagram and API list |
| 5 | UI/UX design | Figma design |
| 6 | Backend development | Login, database and APIs |
| 7 | Frontend development | Pages connected with APIs |
| 8 | AI integration | AI features |
| 9 | Feature completion | Nearly complete system |
| 10 | Testing and bug fixing | Testing report |
| 11 | Deployment | Live website link |
| 12 | Documentation | Final report, user manual and API documentation |
| 13 | Presentation preparation | Presentation slides |
| 14 | Final presentation and viva | Live demo |